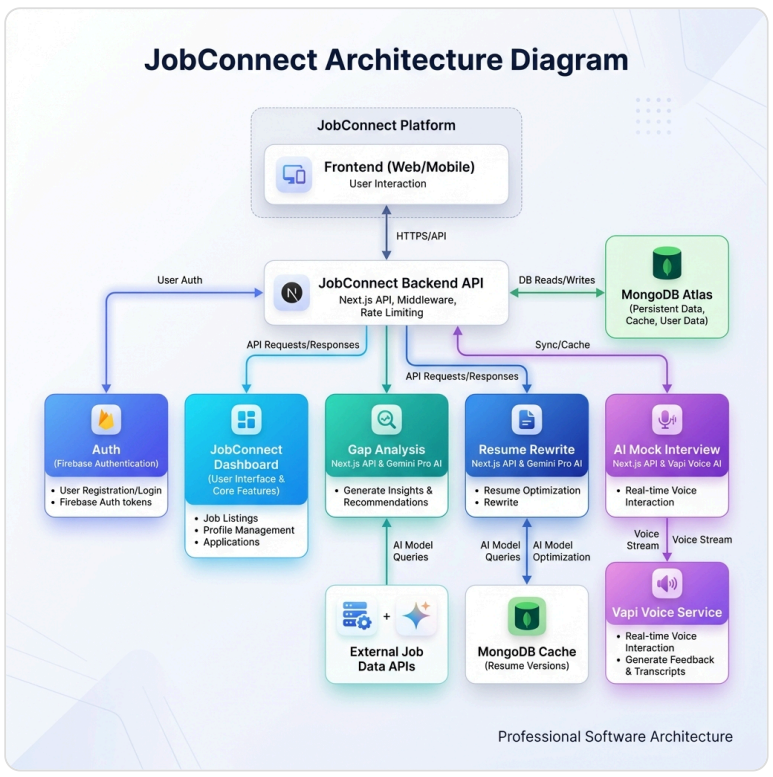
JobConnect: AI-Powered Career Prep Platform

Automated ATS Match Optimization & Real-time Interview Simulation

Objectives

- **Optimize Applications:** Automatically analyze and rewrite candidate resumes to significantly increase ATS (Applicant Tracking System) match rates against specific Job Descriptions.
- **Seamless Document Processing:** Ingest complex user documents (PDFs, DOCX) cleanly and systematically without structural data loss.
- **Actionable Feedback & Skill Mapping:** Provide users with detailed visibility into missing keyword requirements and skill gaps.
- **Realistic Interview Readiness:** Deliver dynamic, generative AI-driven mock interviews based on the exact job requirements matched directly to the user's specific resume context.

System Architecture



Algorithms & Core Parsing Used

NLP (Natural Language Processing)

Core NER (Named Entity Recognition) methodologies to parse, extract, and categorize technical skills, soft skills, and professional experience from unstructured applicant documents.

RAG (Retrieval-Augmented Generation)

Implementation of retrieval augmented mechanisms comparing the analyzed candidate profile (resume) tightly against target role guidelines (job description).

Conclusion

JobConnect successfully bridges the gap between job seekers and automated corporate filtering systems. By leveraging advanced NLP parsing, structural document conversion (Mammoth), and highly constrained Generative AI via RAG architectures, the platform provides scalable and exceptionally accurate career preparation.

Testing indicates a significant improvement in candidate pass rates, raising initial ATS keyword viability scores from an average of ~45% up to ~90% post-analysis.